

Forder — Book IV

Proposition 13

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If AC is a diameter of a circle through B , then ABC is a right angle.

1 Introduction

Forder IV.13 is the theorem that an angle in a semicircle is right. In Forder’s formulation, if AC is a diameter of a circle and B is a third point of that circle, then

$$\angle ABC$$

is a right angle [1].

The statement is familiar, but its formal reconstruction is instructive. The proof does not appeal to a numerical statement saying that the angle is 90° , nor does it identify a semicircle with an angular measure of π . The right-angle conclusion is expressed synthetically: after extending one side of the angle, the two adjacent angles are proved congruent.

The production file is

```
CGJteamLab/Proposition4_13.lean
```

and its public endpoint is

```
hilbert_IV13_diameter_right_angle
```

Unlike IV.12, IV.13 has no large order case split. Its difficulty is different: the formal proof must make explicit the midpoint property of the centre, the two isosceles triangles determined by the radii, the transport of their base angles along the diameter, and the final assembly of two angle decompositions.

2 Classical Configuration

Let K be the centre of the circle and let A, B, C lie on the circle. Assume that A, K, C lie on the diameter carrier, with B off that line.

The source statement uses AC as diameter and concludes that

$$\angle ABC$$

is right. In the production interface the letters are permuted: the endpoints of the diameter are A, B , the third circle point is C , and the target is

$$\angle ACB.$$

This is only a change of names. The geometric content is identical.

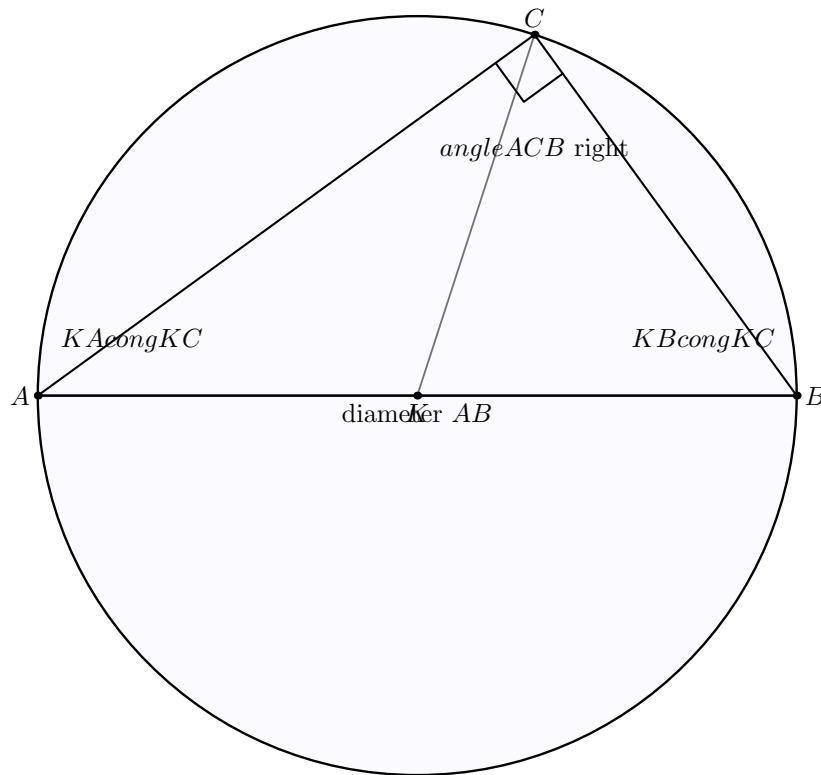


Figure 1: Forder IV.13 in the production naming convention. The centre K lies on the diameter AB , while C is a third point of the circle. The target angle ACB is right.

The central fact hidden in the picture is not merely that K lies on AB . Since A and B are distinct points of one circle centered at K , the centre is the strict midpoint:

$$A - K - B, \quad KA \cong KB.$$

That strict order is used repeatedly to transport angle arms along the same rays.

3 Forder's Statement and Proof

Forder states:

If AC is a diameter of a circle through B , then ABC is a right angle.

His proof begins with the centre O . The three radii are congruent:

$$OA \cong OB \cong OC.$$

Hence the two triangles formed by the centre and the point B are isosceles. Their base angles are equal.

Because the centre lies on the diameter, the ray from A through O is the same ray as the ray from A through the other endpoint of the diameter, and similarly from the opposite endpoint. Thus the equal base-angle information can be read as information about the two pieces of the angle at the circumference.

Forder then extends one side at the circumference and compares the angle inside the triangle with its adjacent exterior angle. The two are shown congruent, which is precisely the synthetic criterion for a right angle [1].

The formal reconstruction follows the same geometry, but expands every order and angle-composition step that the printed proof leaves in the diagram.

4 Formal Statement

The proposition interface lives in `HilbertInterfaceIV.lean`:

```
def HilbertForderIV13DiameterRightAngle
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R A B C : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      H.OnLine K chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      Not (H.OnLine C chord) ->
      HilbertRightAngle Geo A C B
```

The assumptions say:

- A and B are distinct;
- A, B, K lie on one carrier line;
- A, B, C lie on the same circle centered at K ;
- C does not lie on the diameter carrier.

The conclusion

```
HilbertRightAngle Geo A C B
```

is the project’s synthetic right-angle predicate.

The public theorem is:

```
theorem hilbert_IV13_diameter_right_angle
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV13DiameterRightAngle (Geo := Geo)
```

The statement itself needs only incidence and congruence in its interface. The proof is carried out under the full Euclidean-plane structure because it uses the order and angle machinery, including Euclid I.32.

5 The Centre Is the Midpoint of the Diameter

The first substantial step is to upgrade

$$A, K, B \text{ collinear}$$

to the strict order

$$A - K - B.$$

The reusable theorem is

```
hilbert_circle_center_midpoint_of_chord
```

Applied to two distinct circle points A, B whose chord carrier contains the centre, it gives

```
HilbertIsMidpoint Geo K A B
```

and hence in particular

$$\text{Between}(A, K, B).$$

This is more than a convenience. It provides both nondegeneracy and the same-ray information required later:

$$K, B \text{ lie on the same ray from } A,$$

and

$$K, A \text{ lie on the same ray from } B.$$

In the development-stage proof this is immediately unpacked into **Ne** facts, collinearity data, and the reversed betweenness $B - K - A$. The production proof retains the same mathematics even though the reusable midpoint theorem now lives in the interface layer.

6 The Two Isosceles Radius Triangles

Since A, B, C lie on the circle centered at K ,

$$KA \cong KC, \quad KB \cong KC.$$

Thus the triangles KAC and KBC are isosceles.

The proof obtains

$$\angle KAC \cong \angle KCA$$

and

$$\angle KBC \cong \angle KCB$$

from the general isosceles-base-angle theorem.

The diameter order now allows the first arms to be transported.

Because K and B lie on the same ray from A ,

$$\angle KAC = \angle BAC.$$

Because K and A lie on the same ray from B ,

$$\angle KBC = \angle ABC.$$

Therefore the isosceles information becomes

$$\angle BAC \cong \angle KCA, \quad \angle ABC \cong \angle KCB.$$

This is the central comparison inherited directly from Forder's proof.

7 Extending the Side AC

The formal proof next extends AC through C to a point X :

$$A - C - X.$$

This is the line along which the adjacent exterior angle will be formed. The target for rightness is therefore to prove

$$\angle ACB \cong \angle BCX.$$

The production proof does not jump directly to that statement. Instead it applies Euclid I.32 to triangle BAC with AC produced through C to X .

The result supplies a point P between B and X such that

$$\angle ABC \cong \angle BCP$$

and

$$\angle BAC \cong \angle PCX.$$

Geometrically, CP is the interior splitter of the exterior angle $\angle BCX$.

This use of I.32 is one of the main differences between the polished source proof and the formal proof object: the angle-sum relation is represented by an explicit splitter point and two angle-congruence statements.

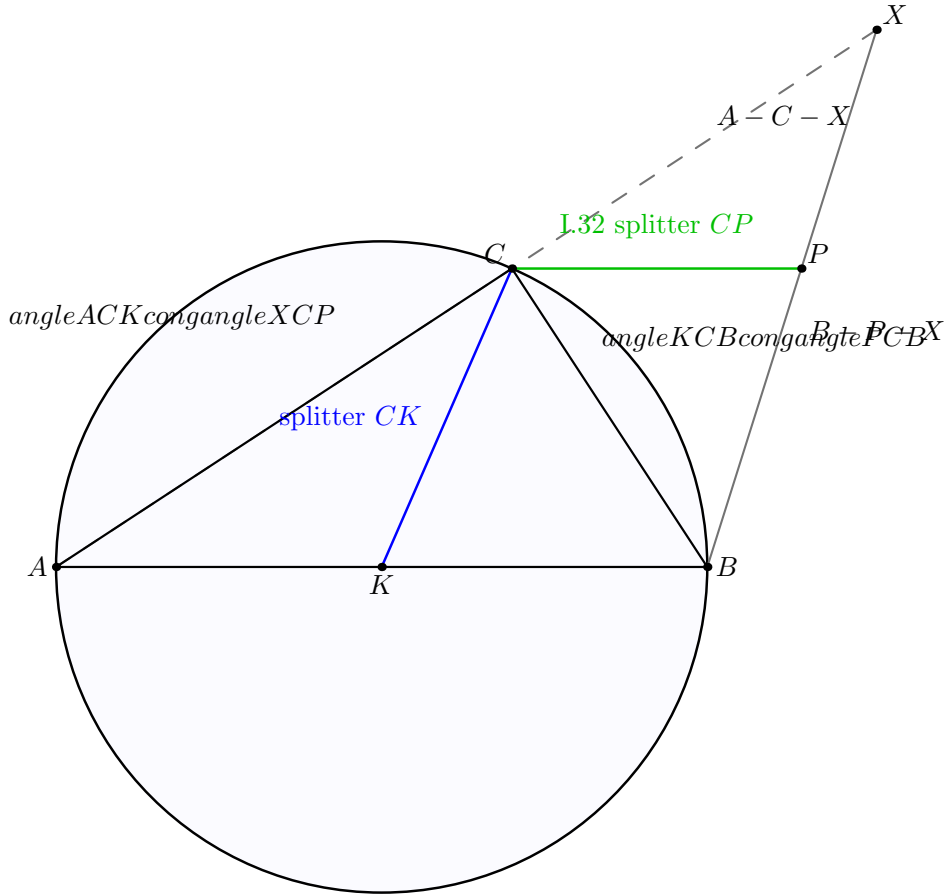


Figure 2: The formal I.32 configuration. The side AC is extended through C to X . The point P lies on BX , and the ray CP splits the exterior angle BCX into the two components corresponding to the remote interior angles of triangle BAC . The ray CK simultaneously splits the original angle ACB .

8 Two Parallel Angle Decompositions

At this stage the proof has two decompositions.

The first is the original angle at C , split by CK :

$$\angle ACB = \angle ACK + \angle KCB$$

in synthetic decomposition form.

The second is the adjacent exterior angle, split by CP :

$$\angle XCB = \angle XCP + \angle PCB.$$

The already established isosceles and I.32 congruences provide the componentwise matches

$$\angle ACK \cong \angle XCP$$

and

$$\angle KCB \cong \angle PCB.$$

The proof then invokes the general angle-addition theorem to conclude

$$\angle ACB \cong \angle XCB.$$

After reversing the order of the arms in the second angle, this becomes

$$\angle ACB \cong \angle BCX.$$

This is exactly the adjacent-angle equality required for a right angle when $A - C - X$.

9 Why the Side-of-Line Work Appears

A conspicuous part of the Lean proof is devoted to carrier lines and same-side/opposite-side conditions for the two decompositions. This is not an accidental formalization overhead.

The general angle-addition theorem does not infer from a diagram that the splitter ray CK lies inside $\angle ACB$, nor that CP lies inside $\angle XCB$. Those facts are certified through plane-separation data.

For the first decomposition, the diameter order $A - K - B$ shows that A and B lie on opposite sides of the carrier CK .

For the second, the I.32 point satisfies $B - P - X$, so B and X lie on opposite sides of the carrier CP .

The proof packages these two facts into the compatibility condition needed by the synthetic angle-addition theorem. Only then may the two component congruences be assembled.

Thus the formal proof makes explicit a fact the source diagram silently supplies: both splitters cut the corresponding angles in the same combinatorial manner.

10 The Final Right-Angle Step

The final result of the angle-addition stage is

$$\angle ACB \cong \angle BCX,$$

with

$$A - C - X.$$

The rays CA and CX are opposite. Hence the angles ACB and BCX are adjacent angles on a straight line. Their congruence is the synthetic definition of rightness used by the project.

The theorem therefore returns

```
HilbertRightAngle Geo A C B
```

without introducing degrees or a numerical angle measure.

This point is worth emphasizing: “right angle” is not proved by evaluating $\angle ACB$ as 90° . It is witnessed geometrically by an extension point X and equality of the two adjacent angles.

11 Lean Formalization

The public API is deliberately small:

```
HilbertForderIV13DiameterRightAngle  
hilbert_IV13_diameter_right_angle
```

The body of the proof depends on reusable geometric results rather than exporting proposition-specific helper lemmas as new API.

Representative reusable ingredients are:

```
hilbert_circle_center_midpoint_of_chord  
hilbert_circle_center_congruent  
hilbert_isosceles_base_angles  
hilbert_sameRay_of_between  
hilbert_angle_eq_of_sameRay_first  
euclid_proposition_32_exterior  
hilbert_angle_addition
```

The proof pattern can be summarized as:

```
have hMidK :  
  HilbertIsMidpoint Geo K A B := ...  
  
have hIsoA :  
  Geo.AngleCongruent K A C K C A := ...  
  
have hIsoB :  
  Geo.AngleCongruent K B C K C B := ...
```

```

rcases HilbertOrder.between_extension A C hAC with
  <X, hACX>

rcases euclid_proposition_32_exterior
  (Geo := Geo) B A C X hBAC hACX
  with <P, hBPX, hABC_BCP, hBAC_PCX>

have hACB_XCB :
  Geo.AngleCongruent A C B X C B :=
  hilbert_angle_addition ...

exact <X, hACX, ...>

```

The production split places the general midpoint and circle lemmas in `HilbertInterfaceIV`. The numbered module therefore owns the mathematical argument of IV.13 without becoming the owner of generic circle infrastructure.

12 Relation to IV.12

Classically, IV.13 can look like an immediate specialization of the inscribed-angle theorem: an angle standing on a diameter intercepts a central straight angle and is therefore half of it.

That is not the dependency chosen by the production formalization. `Proposition4_13.lean` does not import `Proposition4_12.lean`. Instead it proves the right-angle theorem directly from:

- the midpoint order of the centre on a diameter,
- equal radii and isosceles triangles,
- Euclid I.32,
- synthetic angle addition.

This is a mathematically useful independence. IV.13 does not inherit the large addition/subtraction case analysis of IV.12, and IV.14 can subsequently depend on IV.13 through a much narrower theorem.

13 Hilbert and Earlier Euclid Components Used

13.1 Incidence

The proof repeatedly creates carrier lines through two distinct points and uses uniqueness of a line through two points to transfer collinearity and off-line information.

13.2 Order

Betweenness is essential in two places:

$$A - K - B$$

for the diameter midpoint, and

$$A - C - X$$

for the adjacent-angle construction. The I.32 point also gives

$$B - P - X.$$

These order facts generate the same-ray and opposite-side relations needed by angle transport and addition.

13.3 Congruence

Circle membership yields congruent radii. The congruent radii yield isosceles-base-angle congruences. The entire proof remains within segment and angle congruence; no metric coordinate is introduced.

13.4 Euclid I.32

The exterior-angle theorem is the main Euclidean proposition used in the formal assembly. It replaces a numerical equation of angle sums by an explicit point P and two angle-congruence relations.

14 Foundational Boundary

No new axiom is introduced for IV.13. The theorem is proved under

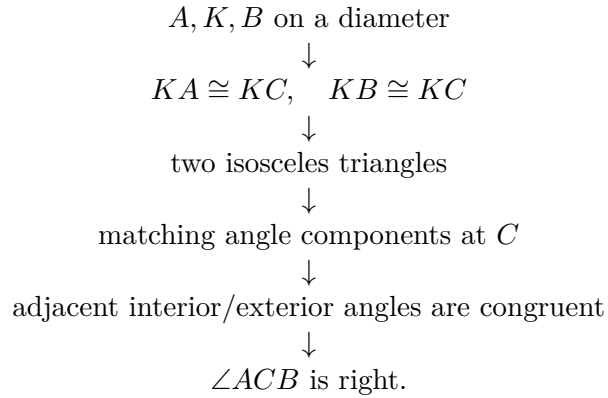
[HE : @HilbertEuclideanPlane Geo H]

using the project’s existing incidence, order, congruence, extension, and Euclidean angle infrastructure.

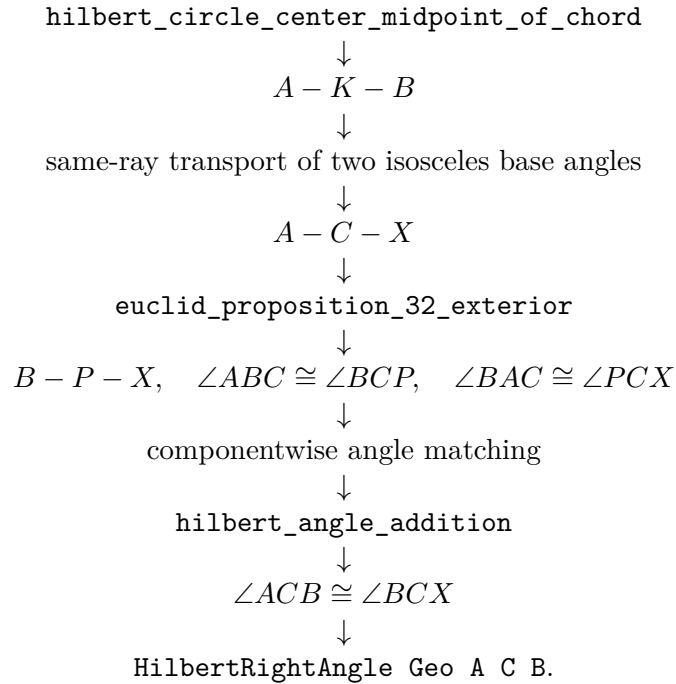
The proof is notably independent of IV.12 as a numbered theorem. Its foundational burden is therefore more transparent than the classical slogan “an angle in a semicircle is half a straight angle” might suggest.

15 Classical and Formal Dependency Structures

The classical argument can be summarized as



The formal proof expands the middle of that chain:



At module level the production dependency is simply

```
HilbertInterfaceIV
-> Proposition4_13
```

16 Diagrammatic Audit

IV.13 needs fewer figures than IV.12, but two figures are justified.

1. The semicircle configuration records the actual theorem and makes the midpoint role of the centre visible.

2. The I.32 configuration explains the formal proof architecture: the splitter CK of the original angle and the splitter CP of the exterior angle are compared component by component.

A separate dependency diagram would add little; the algebraic display above already captures the proof structure.

17 Conclusion

Forder IV.13 is short in the source because the diagram carries much of its order information. The formal reconstruction exposes that information without changing the mathematical idea.

The centre of the diameter is first made into a strict midpoint. Equal radii give two isosceles triangles, and the midpoint order transports their base angles to the two components of the target angle. After extending one side, Euclid I.32 constructs a matching decomposition of the adjacent exterior angle. Synthetic angle addition then shows the two adjacent angles congruent, which is exactly the project's definition of a right angle.

The proposition is also architecturally clean: it depends only on the reusable Book IV interface and does not import IV.12. This makes IV.13 a narrow, independent theorem that can be used directly by IV.14 and later diameter arguments.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.